

Envoy

Envoy — Easy KCNA Notes

What is Envoy?

Envoy is a high-performance proxy that manages and controls network traffic between applications.

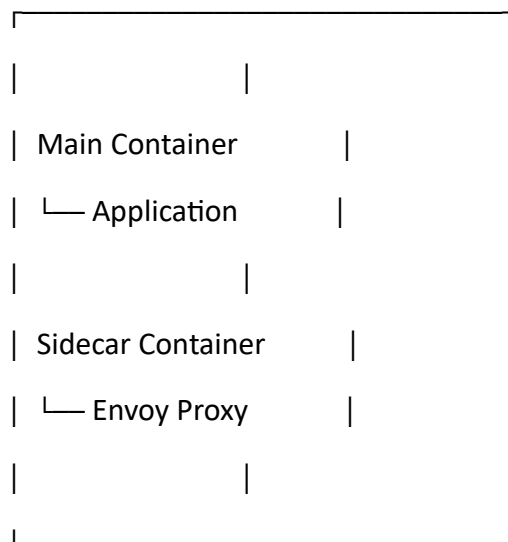
In simple words:

Envoy = A smart traffic helper/proxy for applications.

Envoy with Kubernetes

Envoy is often used as a **sidecar container**.

Pod



The application can send network traffic through Envoy.

Application

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Envoy Proxy

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Other Service

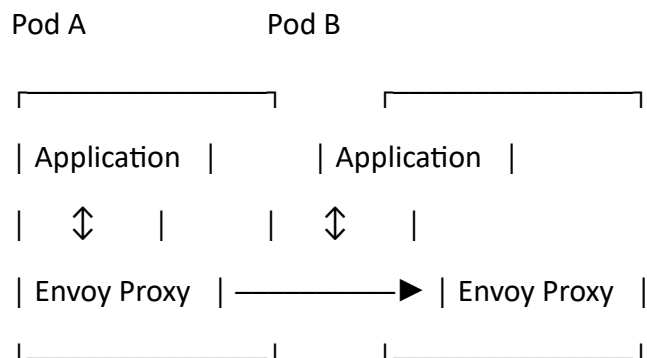
What can Envoy do?

-  Manage network traffic
 -  Route requests to different services
 -  Help with secure communication
 -  Provide traffic information/metrics
 -  Perform health checking
 -  Handle retries and timeouts
 -  Help with load balancing
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Envoy and Service Mesh

Envoy is commonly used as a **data-plane proxy** in a service mesh.

For example:



The applications don't have to handle all the networking themselves. **Envoy proxies can handle many of those communication tasks.**

Envoy vs Sidecar

Don't confuse them:

Sidecar = a design pattern/way of running a helper container alongside the main container.

Envoy = a specific **proxy software** that can be used as that sidecar.

Sidecar



Design pattern

Envoy



Proxy software

 **KCNA Memory Trick**

Envoy = Traffic Police 

It stands between services and helps control, route, and observe their network traffic.

One-line exam definition:

Envoy is a high-performance proxy commonly deployed as a sidecar to manage communication between services.